

Cristiano Ronaldo: Career Analysis Report

A deep dive into Ronaldo's goal-scoring patterns using **Univariate Analysis**, **Bivariate Analysis**, and **Career Analysis**

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Project Objective



Problem Statement

Analyze Cristiano Ronaldo's goal-scoring patterns using statistical and visual techniques, focusing on:

- Goal distributions and spread
- Relationships between variables
- Performance evolution over time



UNIVARIATE ANALYSIS

Goal Minute Distribution

Univariate analysis studies one variable at a time – here, the timing of each goal across a match.

- ① Most goals are concentrated within certain match periods, with some unusual timings appearing as **outliers** — revealing Ronaldo's scoring behavior throughout matches.

Penalty vs. Goal Minute



What the Data Shows

A boxplot comparing the timing of **penalty** and **non-penalty** goals reveals:

- Variation in goal timings across both types
- Outliers representing unusually early or late goals
- Distinct scoring patterns between penalty and open-play goals

Goals Per Year

This line graph tracks Ronaldo's annual goal tally, identifying **peak scoring seasons** and relative dips in output.



Higher points indicate standout seasons, while lower points reflect injury, transition, or adaptation periods.

Own Goal vs. Goal Minute

Boxplot Insights

Comparing the timing of **own goals** and **normal goals** involving Ronaldo highlights:

- Variations in goal timing distributions
- Unusual scoring patterns as outliers
- Differences in match-phase concentration

📄 Bivariate analysis revealed key relationships between variables, including yearly trends and timing patterns.





Bivariate Analysis: Key Takeaways

Yearly Trends

Clear peaks and troughs in annual goal output identified

Timing Patterns

Penalty and open-play goals cluster in distinct match phases

Scoring Behavior

Outliers reveal unusual but recurring scoring moments

Career Timeline

Career analysis tracks how Ronaldo's performance **evolved over time** — identifying growth phases, peak years, and fluctuations.



Penalty Goals Per Year



Annual Penalty Performance

A bar chart of penalty goals per year shows:

- Higher bars indicate **stronger penalty seasons**
- Fluctuations tied to team tactics and competition type
- Consistent penalty conversion across his career

✓ Penalty goals form a significant and reliable portion of Ronaldo's annual totals.

Final Conclusion

The analysis successfully uncovered Ronaldo's goal-scoring story through three analytical lenses.



Univariate Analysis

Revealed goal timing distributions, central tendency, and outliers within matches



Bivariate Analysis

Established relationships between penalty types, own goals, and yearly output



Career Analysis

Highlighted long-term trends, transformation periods, and peak performance years

-Thank You